

CE 310

Week 11 — Hypothesis Testing

t-Tests, ANOVA, and Effect Size

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The Question Behind the Numbers

Weeks 9–10 showed that occupied hours average **38.9 kWh/hr** of cooling and unoccupied hours average **10.6 kWh/hr** — a difference of 28.3 kWh/hr.

But how do you know that difference isn't just sampling noise?

What if the building had a particularly busy stretch of summer afternoons in this dataset that happened to inflate the occupied mean? What if a different year of data would show only a 5 kWh gap?

Hypothesis testing is the formal answer:

- It asks: *what is the probability that sampling noise alone produced a gap this large, if the true means were actually equal?*
- If that probability is very small ($p < 0.05$), we conclude the gap is **real** — not noise.

Anatomy of a Hypothesis Test

Every test follows the same five steps — memorize the structure once; apply it to any test.

1. **State H_0 (null):** the default "no effect" assumption, always written with $=$. Example: $H_0 :$

$$\mu_{occ} = \mu_{unocc}$$

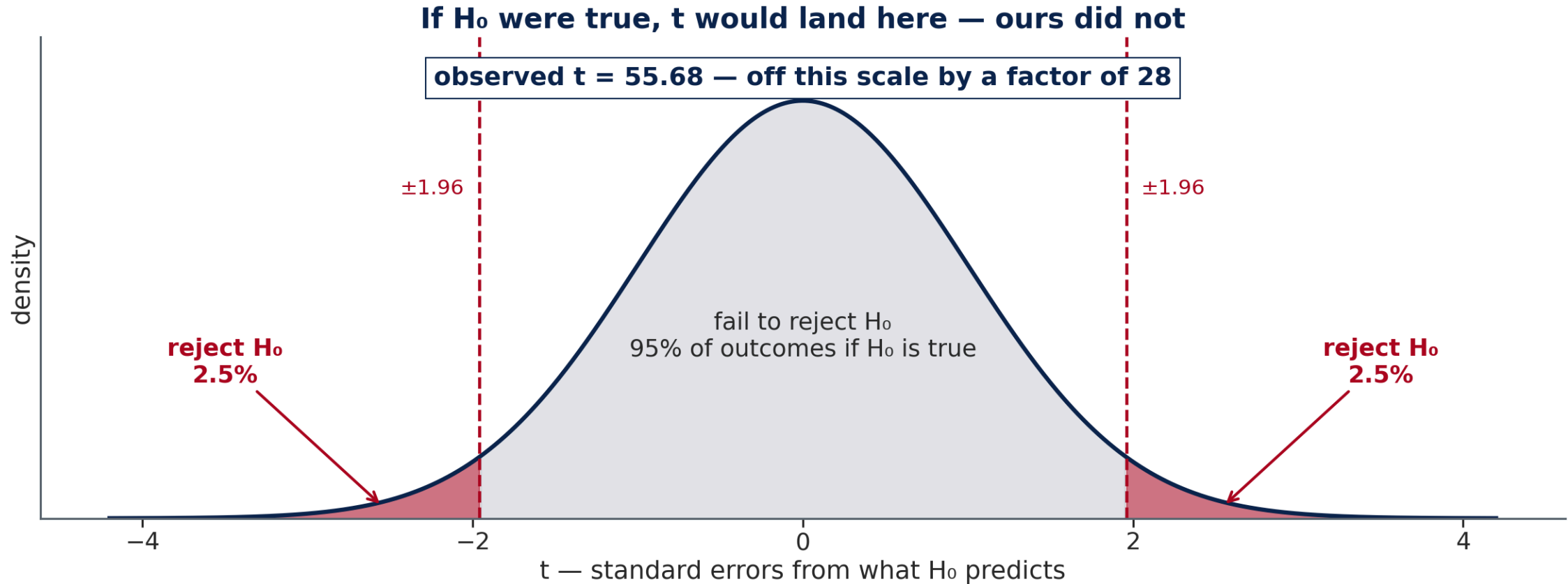
2. **State H_1 (alternative):** what you want to show. *Two-tailed* ($\neq$): difference in either direction — most common. *One-tailed* ($>$ or $<$): specific direction, requires prior justification.
3. **Choose α — before seeing the data:** typically 0.05 (5% false-positive rate accepted in advance). Choosing α after seeing the result is p-hacking.

Anatomy of a Hypothesis Test (cont.)

4. **Compute the test statistic:** measures deviation from H_0 in standard-error units. Two groups $\rightarrow$ t-statistic. Three or more groups $\rightarrow$ F-ratio.
5. **Interpret the p-value:** probability of a result this extreme or more extreme *if H_0 were true*. $p < \alpha$ $\rightarrow$ reject H_0 .

"Fail to reject H_0 " $\neq$ " H_0 is true." The data simply lack enough evidence to rule it out — like a "not guilty" verdict that does not mean innocent.

What "If H_0 Were True" Actually Means



Every hypothesis test compares one observed number against the range that number *would* take if H_0 were true. Everything else is bookkeeping.

5 Steps Applied — Occupied vs. Unoccupied Cooling

The five-step structure

1. H_0 — null assumption (always uses $=$)
2. H_1 — what to show ($\neq$, $>$, or $<$)
3. α — false-positive rate, set in advance
4. **Test statistic** — signal $\div$ noise in SE units
5. **Decision** — compare p-value to α

Applied to this dataset (n = 8,760)

1. $H_0 : \mu_{occ} = \mu_{unocc}$ — occupancy has no effect
2. $H_1 : \mu_{occ} \neq \mu_{unocc}$ — two-tailed
3. $\alpha = 0.05$
4. $t \approx 55.7$ (gap of 28.3 kWh $\div$ SE ≈ 0.51)
5. $p \approx 0 < 0.05 \rightarrow$ **Reject H_0**

The same five-step template applies to every test this week: the one-sample benchmark t-test, the one-way ANOVA, and every coefficient in the $P > |t|$ regression output.

Two Ways a Hypothesis Test Can Be Wrong

Rejecting or failing to reject H_0 is a decision made under uncertainty — it can be wrong in two distinct ways.

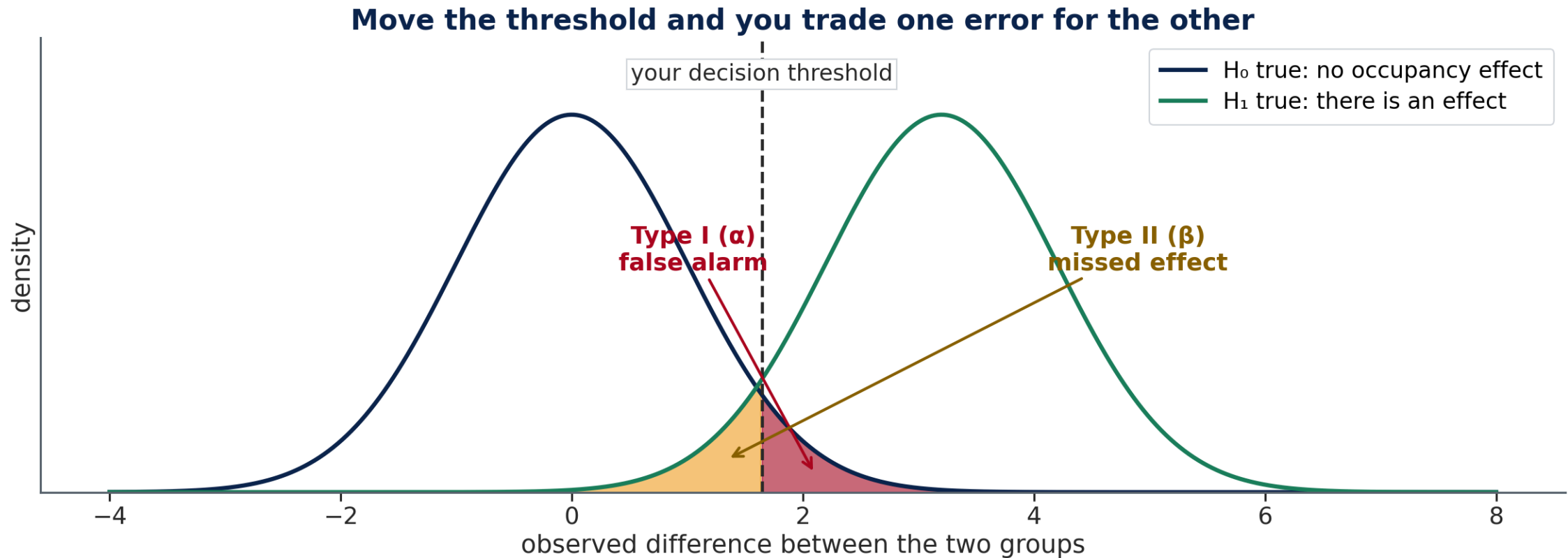
	H_0 is actually true	H_0 is actually false
You reject H_0	Type I error (false positive) — probability = α	Correct decision
You fail to reject H_0	Correct decision	Type II error (false negative)

Two Ways a Hypothesis Test Can Be Wrong (cont.)

- **Type I error (α):** you conclude there's an effect when there isn't one. This is exactly the "false alarm" rate from Step 3 of the 5-step framework — choosing $\alpha = 0.05$ means you accept a 5% chance of a false positive on any single test.
- **Type II error:** you conclude there's no effect when there actually is one — often from too small a sample or too small a true effect relative to noise.

💡 α is not a mistake rate you observe after the fact — it's a false-positive rate you accept in advance. The "26% false alarm rate across 6 t-tests" result later this lecture is Type I error compounding across multiple tests.

The Two Errors Trade Against Each Other



You cannot shrink both at once with fixed data. Moving the threshold left catches more real effects and raises the false-alarm rate; moving it right does the reverse.

The t-Statistic — What It Measures

The t-statistic answers: *how many standard errors is the observed difference from what H_0 predicts?*

$t = (\text{mean}_1 - \text{mean}_2) / \text{standard error of the difference}$

- **Numerator:** the observed difference between group means
- **Denominator:** how much that difference varies due to chance (shrinks as n grows)

Interpreting t :

- $|t|$ close to 0 $\rightarrow$ difference is within sampling noise $\rightarrow$ do not reject H_0
- $|t|$ large ($> \sim 2$) $\rightarrow$ difference is many standard errors from zero $\rightarrow$ reject H_0

For BH cooling: **$t \approx 55.7$** — the 28.3 kWh gap is 55.7 standard errors from zero. Sampling noise cannot produce a t-statistic this large.

Two-Sample t-Test — Occupied vs. Unoccupied Cooling

H₀: Mean cooling during occupied hours = mean cooling during unoccupied hours

H₁: The means are not equal · **α = 0.05**

A **two-sample t-test** compares the occupied and unoccupied means and returns the *t* statistic and its *p*-value.

Occupied mean: **38.9 kWh/hr** (n = 2,600) · Unoccupied mean: **10.6 kWh/hr** (n = 6,160)

Expected: $t \approx 55.7$ · $p \approx 0.0 \rightarrow$ **Reject H₀**

The *p*-value is effectively zero — Python rounds to `0.0000e+00`. With $t = 55.7$, the 28.3 kWh gap is 55.7 standard errors from zero. Sampling noise cannot explain a difference this large.

A Caution: These Data Are Not Independent

$t = 55.7$ assumes all 8,760 observations are independent. **Hourly weather data is not.**

Autocorrelation: The cooling demand at 2 PM depends on conditions at 1 PM — same outdoor temperature trend, same building thermal mass, same occupancy ramp. Consecutive hours share information.

What this means for the p-value:

- Correlated data has **effective n** far smaller than 8,760
- For hourly building data, effective $n \approx 300$ –1,000
- True t-statistic would be smaller — but for a 28 kWh gap, still enormous

Does the conclusion change here?

No — the 28.3 kWh/hr difference between occupied and unoccupied cooling is so large it would dominate even with effective $n = 50$.

One-Sample t-Test — Comparing to a Benchmark

When you want to test whether a sample mean equals a **known reference value** (not another sample), use the one-sample t-test.

Example: An energy benchmark sets the expected average cooling intensity at **20 kWh/hr** for this building type. Does the dataset match the benchmark?

H₀: $\mu_{\text{cooling}} = 20 \text{ kWh/hr}$

H₁: $\mu_{\text{cooling}} \neq 20 \text{ kWh/hr}$

A **one-sample t-test** compares the observed mean against a fixed benchmark — here the 20 kWh/hr design figure.

The annual mean in this dataset is **19.0 kWh/hr** — only 1 kWh below benchmark. But with $n = 8,760$ and a small standard error, even a 1 kWh deviation can be statistically significant.

This introduces the central tension of Week 11: **statistical significance $\neq$ practical significance.**

Quick Check — Big Sample, Big Problem?

60 seconds · pick A or B

Three comparisons all have $p \approx 0.000$ with $n = 8,760$:

- Cooling: occupied vs. unoccupied — $\Delta = 28.3 \text{ kWh/hr}$
- Cooling: weekday vs. weekend — $\Delta = 2.1 \text{ kWh/hr}$
- Outdoor temp: occupied vs. unoccupied — $\Delta = 3.2^\circ\text{F}$

All three are "statistically significant." Which interpretation is correct?

- **A)** All three effects matter equally for HVAC design — same p-value means same importance
- **B)** The p-value tells you the effect is real, not how large it is

✓ Answer: B — p-values Don't Measure Size

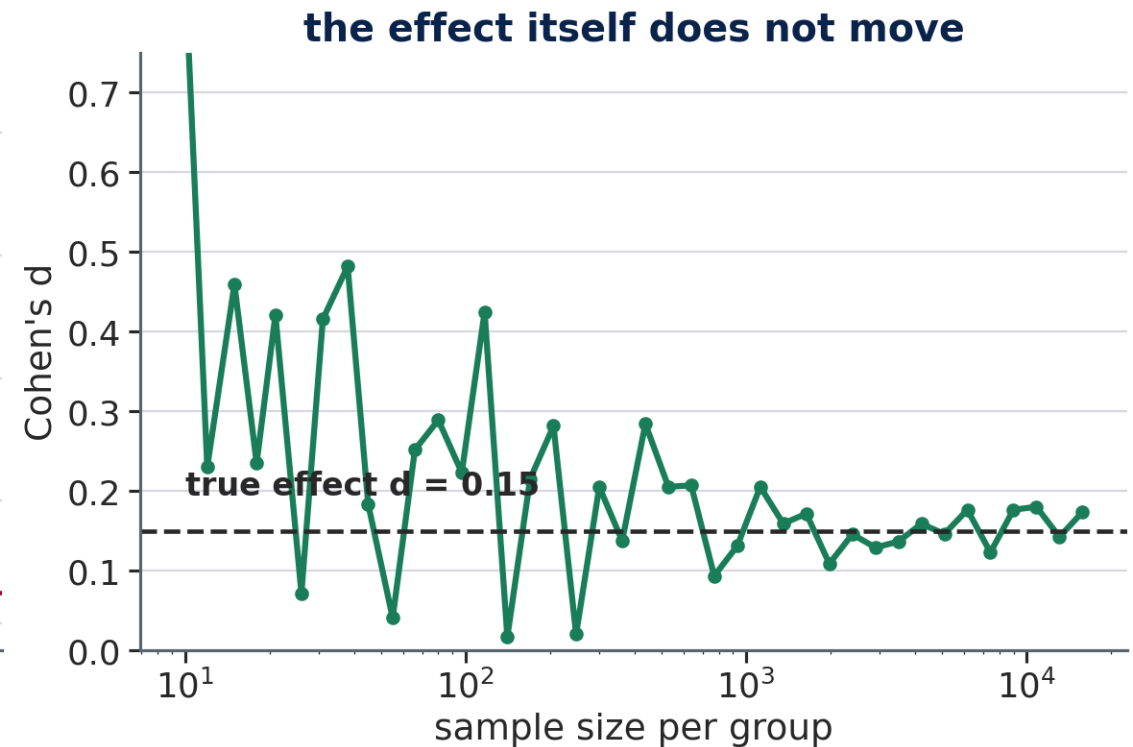
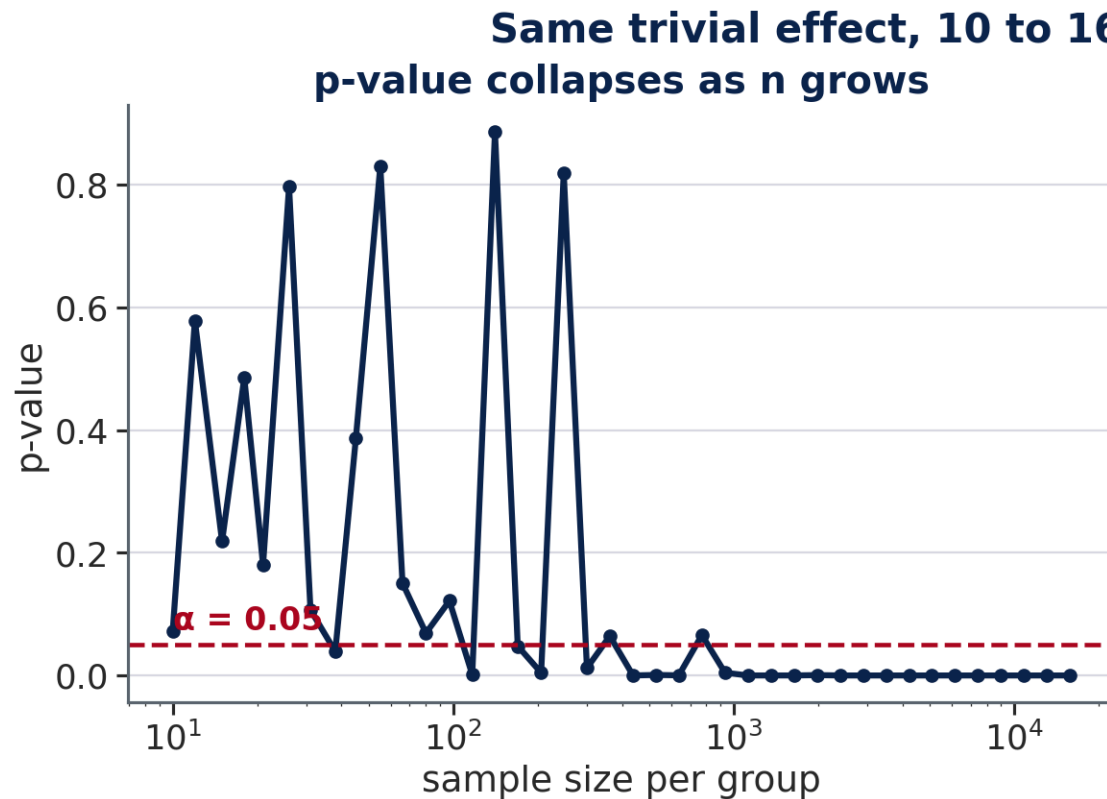
Statistical significance $\neq$ practical significance

Comparison	p-value	Effect size	Engineering implication
Cooling: BH vs. no-BH	≈ 0.000	28.3 kWh/hr	Drives equipment sizing — use it
Cooling: weekday vs. weekend	≈ 0.000	2.1 kWh/hr	Real but too small to act on
Outdoor T: BH vs. no-BH	≈ 0.000	3.2°F	Statistically clear; operationally minor

The rule: with $n > 5,000$, nearly every difference is statistically significant. A significant p-value means the effect is **real** — it does NOT mean the effect is **large enough to matter**.

⚠ Always report effect size alongside p-value when n is large. In engineering, the question is never "is this real?" — it's "is this large enough to change my decision?"

Same Effect, Different n



The effect never changed. Only our ability to detect it did. This is why a p-value alone cannot tell an engineer whether something is worth acting on.

Statistical vs. Practical Significance

With large datasets, almost every difference is statistically significant. A p-value tells you the effect is **real** — not noise. It does not tell you whether the effect is **large enough to matter**.

Example from the dataset:

Result	p-value	Conclusion
Cooling: occupied vs. unoccupied	≈ 0.000	Statistically significant
Cooling: weekday vs. weekend	≈ 0.000	Statistically significant
Outdoor temp: occupied vs. unoccupied	≈ 0.000	Statistically significant

⚠ Always report effect size alongside p-values when n is large. Statistical significance is necessary but not sufficient for an engineering decision.

Effect Size — Cohen's d

Cohen's d measures a difference in standard-deviation units — independent of sample size.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}$$

Conventional benchmarks:

| |d| | Classification |

|---|---|

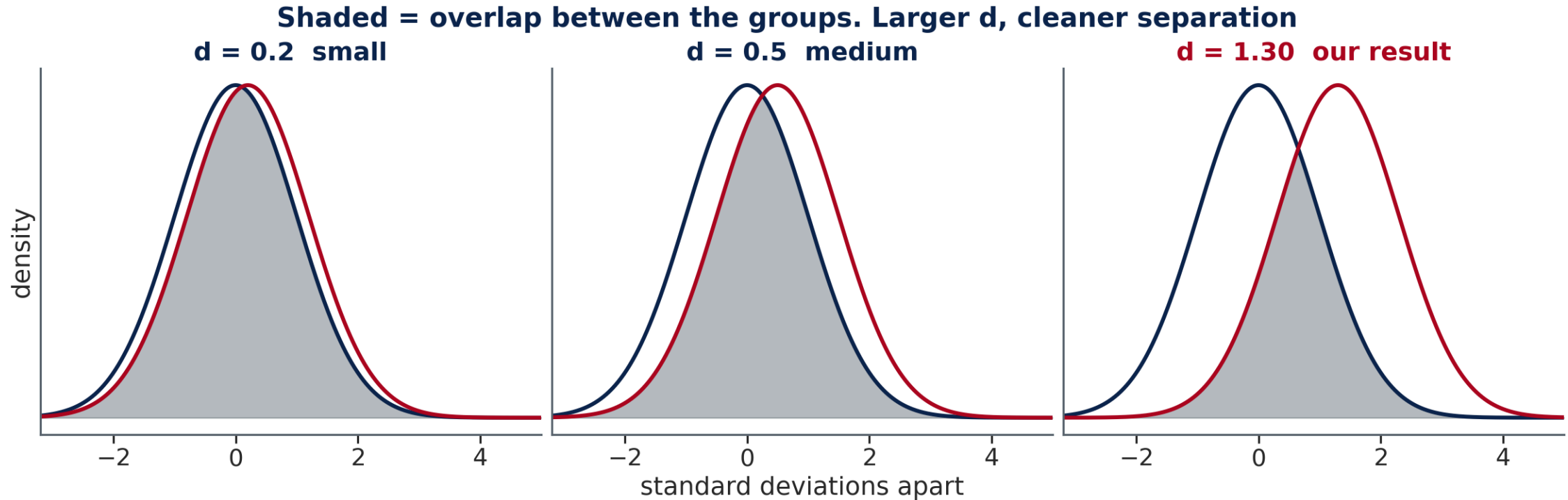
| 0.2 – 0.5 | Small |

| 0.5 – 0.8 | Medium |

| > 0.8 | Large |

- Cooling, BH vs. no-BH: $d \approx \mathbf{1.30}$ (large — occupancy dominates cooling load)
- Cooling, weekday vs. weekend: $d \approx \mathbf{0.38}$ (small — DayType captures less of the signal)

What Cohen's d Looks Like



Our occupancy effect is the right-hand panel: the two groups barely overlap. That is what " $d = 1.30$, large" means in pictures rather than in a table.

Quick Check — Multiple t-Tests

60 seconds · pick A or B

You want to compare cooling across all four seasons. Instead of ANOVA, you run **6 pairwise t-tests** (all season pairs), each at $\alpha = 0.05$.

What is the actual probability of at least one false alarm across all 6 tests?

- **A)** 5% — same as each individual test
- **B)** Something higher — multiple tests inflate the error rate

✓ False Alarm Rate: $\approx 26\%$

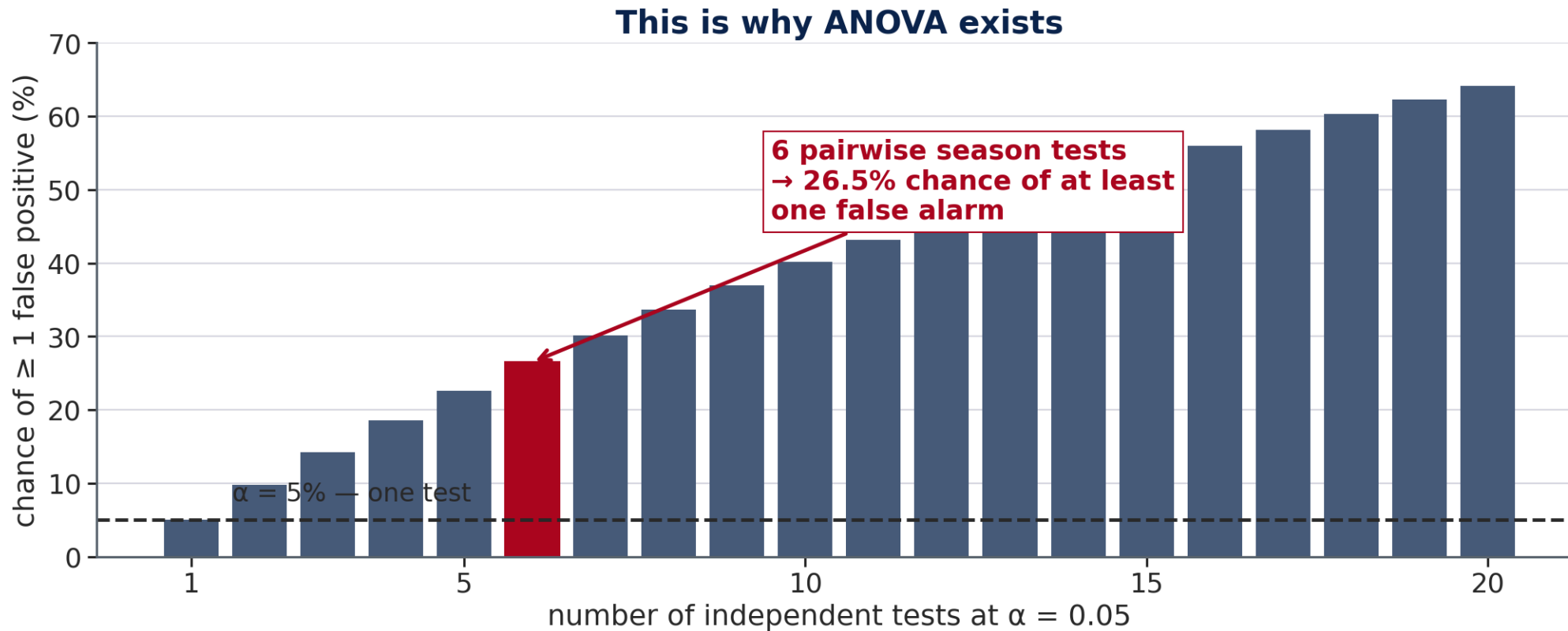
Running 6 t-tests at $\alpha = 0.05$ is not $\alpha = 0.05$

$$P(\text{at least one false alarm}) = 1 - (1 - \alpha)^k = 1 - (0.95)^6 \approx \mathbf{26\%}$$

Tests run	Individual α	Family-wise false-alarm rate
1 test	5%	5%
3 tests	5%	14%
6 tests	5%	26%
10 tests	5%	40%

The fix: use **ANOVA** to first test "does any season differ?" (one test, one $\alpha = 0.05$). If significant, use **Tukey HSD** for pairwise comparisons.

False Alarms Compound



Each test is honest on its own. Run enough of them and a false positive somewhere becomes the expected outcome rather than the exception.

One-Way ANOVA — Comparing Three or More Groups

The t-test compares **two** groups. When there are $k \geq 3$ groups (e.g., four seasons), use ANOVA.

H_0 : All group means are equal ($\mu_{\text{Fall}} = \mu_{\text{Spring}} = \mu_{\text{Summer}} = \mu_{\text{Winter}}$)

H_1 : At least one group mean differs

One-way ANOVA asks a single question of all four seasons at once: is *any* seasonal mean different from the others?

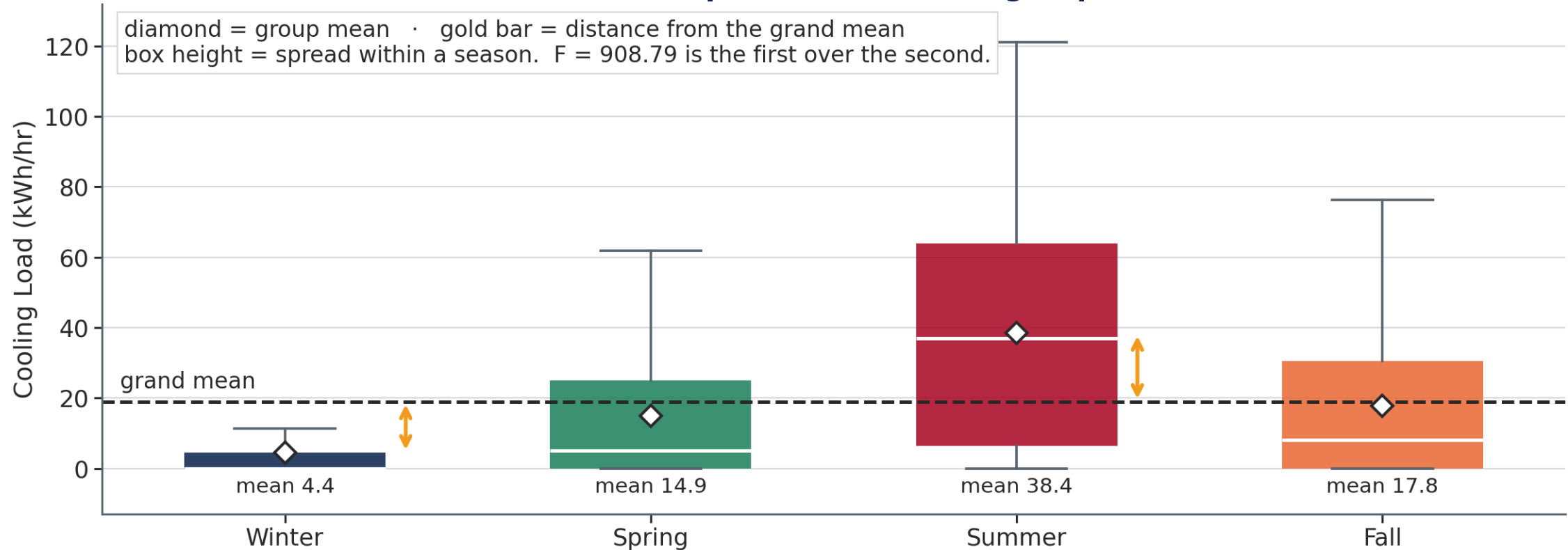
The **F-ratio** = between-group variance / within-group variance:

- Large $F \rightarrow$ the groups are far apart relative to the spread within each group $\rightarrow$ reject H_0
- $F \approx 1 \rightarrow$ the group means don't deviate more than within-group noise $\rightarrow$ do not reject H_0

Important limitation: ANOVA tells you *that* at least one group differs — not *which* groups. You need a **post-hoc test** to identify which specific pairs are different.

What ANOVA Is Comparing

ANOVA asks one question of all four groups at once



F is a ratio: how far apart the group means are, divided by how spread out the data is inside each group. $F = 908.79$ says the seasonal signal dwarfs the hour-to-hour noise.

Post-Hoc Testing — Tukey HSD

After rejecting the ANOVA null hypothesis, **Tukey's Honestly Significant Difference (HSD)** test compares all pairs of groups while controlling the family-wise error rate.

Tukey's HSD follows the ANOVA and reports every pairwise comparison with its confidence interval, already corrected for the number of comparisons made.

The tutorial covers running it and reading the table.

The output table has one row per pair (e.g., Fall–Spring, Fall–Summer, ...). Each row shows:

- `meandiff` — the estimated difference between group means
- `reject` — True if that pair is significantly different at $\alpha = 0.05$

Why not just run all six pairwise t-tests? Running multiple tests inflates the probability of a false alarm. With 6 tests at $\alpha = 0.05$, there is a 26% chance of at least one false positive. Tukey HSD adjusts for this.

Connection to Regression — t-Values in OLS Output

The `P>|t|` column in the `statsmodels` summary is doing a t-test on every coefficient:

H_0 : that coefficient = 0 (the predictor has no linear effect on the outcome)

H_1 : the coefficient $\neq 0$

The regression output already carries a t statistic and p -value for **each coefficient** — a hypothesis test per predictor, computed when the model was fit.

For Model 2: BH t-stat ≈ 55.7 — the same value you get from `ttest_ind(occ, unocc)` when outdoor temperature is not controlled. The regression t-stat is actually a **conditional** version: it tests whether BH is a real predictor **after** removing the temperature effect.

This is why the regression BH coefficient (17.9 kWh/hr) is smaller than the raw group difference (28.3 kWh/hr) — the regression t-test is testing a different, cleaner question.

When to Use Which Test

Situation	Test	Python function
Compare two group means	Two-sample t-test	<code>ttest_ind(g1, g2)</code>
Test a mean against a known value	One-sample t-test	<code>ttest_1samp(data, val)</code>
Compare three or more group means	One-way ANOVA	<code>f_oneway(g1, g2, g3, ...)</code>
Find which pairs differ after ANOVA	Tukey HSD	<code>pairwise_tukeyhsd(...)</code>
Measure practical size of an effect	Cohen's d	compute manually

Workflow: ANOVA first (does any group differ?) → if rejected, Tukey HSD (which pairs differ?) → report effect size (how large is the effect?).

Bridge to Machine Learning

Concept	Statistics	ML equivalent
Null hypothesis test	"Is this predictor's coefficient zero?"	Feature importance test
p-value threshold ($\alpha = 0.05$)	Significance cutoff	Feature selection filter
Cohen's d	Effect size — standardized difference	Normalized feature contribution
ANOVA F-test	Does feature explain group variance?	Between-class variance ratio

In ML, **feature selection** often uses statistical tests under the hood — for example, selecting features by ANOVA F-score before feeding them to a model. The logic is identical to what you did today: does this variable explain meaningful variance across groups?

Looking Ahead — Week 12: Sensitivity Analysis

This week: *is an observed difference statistically real?*

Next week: *if model inputs are uncertain, how much uncertainty does that create in the output?*

Sensitivity analysis identifies which input variable drives the most output variation. You'll apply it to the regression model from Weeks 9–10:

- How much does uncertainty in outdoor temperature affect predicted annual cooling?
- How much does the occupancy schedule assumption matter?
- Which input deserves the most careful measurement?

This is a core engineering skill: knowing where to invest effort in reducing uncertainty.